

# Module Summary: Iris Data Workflow

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Project: Iris Data Workflow (data\_workflow.ipynb)

## Overview

This project builds a complete exploratory data workflow around the classic Iris Plants Dataset, implemented in the Jupyter notebook `data_workflow.ipynb`. Using `pandas`, `NumPy`, `Matplotlib`, and `Seaborn`, the notebook ingests the raw data, applies reusable cleaning functions, and analyzes the four floral measurements collected for three iris species. The dataset is the Iris Plants Dataset, publicly available from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/53/iris>). The workflow is designed to be reproducible: every step, from raw data to final figures, runs end-to-end from a single notebook and a pinned `requirements.txt` file.

## Dataset Description

The Iris Plants Dataset, originally collected by Fisher (1936), contains 150 rows and 5 columns: four continuous measurements in centimeters (sepal length, sepal width, petal length, petal width) and one categorical label (species), with 50 samples for each of three species -- Setosa, Versicolor, and Virginica. This project focused primarily on petal length and petal width, which the correlation matrix computed in this project (Figure 3) shows to be the measurements most strongly associated with species identity. Sepal width was also examined because it behaved differently from the other three measurements, showing weak negative correlations rather than the strong positive correlations seen among the other variables. The dataset contains no missing values by design, which made it a good candidate for demonstrating a full cleaning-to-visualization workflow without the complications of large-scale missing data.

## Workflow Description

**Ingestion:** The raw `bezdekIris.data` file (comma-separated, no header) was converted into `iris.csv` with an explicit header row, then loaded with `pandas.read_csv` into a `DataFrame`.

**Cleaning:** Three functions were written to standardize species labels (removing the 'Iris-' prefix), remove duplicate or missing rows, and flag statistical outliers using a mean plus/minus  $n$ -standard-deviations rule.

**Exploratory analysis:** Three functions computed descriptive statistics, per-species group means and standard deviations, and a Pearson correlation matrix across the four measurements.

Visualizations: Three plots were produced -- a histogram of petal length by species, a scatter plot of sepal length versus petal length by species, and a correlation heatmap of all four measurements.

Summary: A markdown section in the notebook interprets the statistical and visual findings in plain language, complementing this report.

## **Key Decisions and Assumptions**

Cleaning choices: Duplicate and missing-value removal was applied dataset-wide rather than per species, and outliers were flagged (not deleted) using a 3-standard-deviation threshold on the full sample. Flagging rather than removing outliers keeps the cleaning step transparent and auditable rather than silently altering the dataset, which is consistent with established guidance on documenting and editing data abnormalities rather than discarding them outright (Van den Broeck et al., 2005). Because the Iris dataset is small, well-curated, and free of missing values, aggressive cleaning was unnecessary; the cleaning functions were instead designed to be defensive and reusable rather than dataset-specific.

EDA focus: descriptive statistics and correlations were computed across the whole dataset and by species group, because the central question in this dataset is how measurements differ between species.

Visualization design: each plot targets a specific question -- the histogram (Figure 1) shows within-feature separability by species, the scatter plot (Figure 2) shows the joint relationship between two measurements, and the heatmap (Figure 3) summarizes redundancy and correlation across all four measurements. Structuring the workflow into named, documented functions rather than inline, one-off code, and executing the whole notebook top-to-bottom, follows the same ingestion-cleaning-EDA-visualization structure promoted in Danchev's (2022) open, reproducible data science curriculum, which was used as a direct model for how this notebook's workflow was organized.

## **Results and Interpretation**

Figure 1 shows that Setosa petal lengths form a distinct, non-overlapping cluster well below those of Versicolor and Virginica, while Versicolor and Virginica overlap partially. Figure 2 reinforces this pattern: Setosa separates cleanly from the other two species on sepal length versus petal length, while Versicolor and Virginica form two elongated, partially overlapping clouds. Figure 3's correlation heatmap shows a very strong positive correlation between petal length and petal width (approximately 0.96) and moderately strong positive correlations between petal measurements and sepal length, while sepal width correlates weakly and negatively with the other three measurements. Together, these results indicate that petal measurements, not sepal measurements, carry most of the information needed to distinguish species in this dataset, consistent with the original findings reported for this dataset (Fisher, 1936).

## **Responsible Practice (Bias and Data Quality)**

Even in a small, well-known teaching dataset, cleaning choices can introduce bias. Applying a single outlier threshold across the whole dataset, rather than within each species group, could disproportionately flag legitimate values in species with naturally wider spread (for example, Virginica), which would bias downstream analysis against that group. Likewise, removing duplicate rows dataset-wide assumes that identical measurement rows are true duplicates rather than two flowers that legitimately share the same measurements; on a larger or noisier dataset, this assumption could silently discard valid data. Because data cleaning decisions can materially change study conclusions, best practice calls for documenting cleaning rules and keeping them auditable and reversible (Van den Broeck et al., 2005), which is why this project's cleaning functions flag outliers in a new column instead of deleting rows outright. A useful next step to further reduce this risk would be to re-run outlier detection within each species subgroup and to report how many rows each cleaning step actually affects.

## **Reproducibility**

Anyone can reproduce this project by cloning the repository, installing the pinned dependencies with 'pip install -r requirements.txt', and opening data\_workflow.ipynb with 'jupyter notebook' (or executing it non-interactively with 'python -m nbconvert --to notebook --execute data\_workflow.ipynb'). Because iris.csv, requirements.txt, and the notebook are all version-controlled together in the project's Git repository, the exact code, data, and dependency versions used to produce these results are captured alongside the results themselves. Work was committed to Git incrementally as each task was completed -- data loading, cleaning, exploratory analysis, visualization, and documentation -- which keeps the history of decisions inspectable and allows any step to be reviewed or reverted independently. This mirrors the emphasis on notebook-based, dependency-pinned, end-to-end reproducible workflows described in Danchev's (2022) open learning resource on reproducible data science with Python, which directly informed the ingestion-through-documentation structure used here.

## Sources and Citations

Danchev, V. (2022). Reproducible data science with Python: An open learning resource. *Journal of Open Source Education*, 5(56), 156. <https://doi.org/10.21105/jose.00156>

Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2), 179-188. <https://doi.org/10.1111/j.1469-1809.1936.tb02137.x>

Van den Broeck, J., Argeseanu Cunningham, S., Eeckels, R., & Herbst, K. (2005). Data cleaning: Detecting, diagnosing, and editing data abnormalities. *PLoS Medicine*, 2(10), e267. <https://doi.org/10.1371/journal.pmed.0020267>